Shuffling and RSK at $q = 0$ produce different conditional step distributions

Verification note

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1 Setup

We compare two algorithms that produce uniform random domino tilings of the Aztec diamond $\mathcal{A}_n$:

EKLP Shuffling. Builds tilings iteratively via $\mathcal{A}_0 \to \mathcal{A}_1 \to \cdots \to \mathcal{A}_n$. Each step $\mathcal{A}_k \to \mathcal{A}_{k+1}$ consists of three phases:

1. *Delete:* Find and remove all "bad blocks" — colliding domino pairs (N-above-S or E-left-of-W in the same 2×2 block).
2. *Slide:* Move each remaining domino one unit in its natural direction (N↑, S↓, E→, W←).
3. *Fill:* Each empty 2×2 block in $\mathcal{A}_{k+1}$ is independently filled with a horizontal or vertical pair, probability $\frac{1}{2}$ each.

RSK Growth Diagram at $q = 0$. Fills an $(n+1) \times (n+1)$ staircase grid of partitions. Each cell (i, j) receives an independent fair Bernoulli bit, and the partition at (i, j) is determined by the VH bijection (which is deterministic at $q = 0$). The boundary of the sub-staircase at level k encodes a tiling of $\mathcal{A}_k$; the boundary layer from level k to $k+1$ consists of exactly $k+1$ cells, hence $k+1$ random bits.

Both algorithms produce the correct uniform distribution on domino tilings. The question is whether their *conditional* distributions for individual growth steps $\mathcal{A}_k \to \mathcal{A}_{k+1}$ coincide.

2 Observation: the conditional distributions differ

Proposition 1. *The EKLP shuffling and the RSK growth diagram at $q = 0$ produce different conditional distributions* $\Pr(\textit{tiling of } \mathcal{A}_{k+1} \mid \textit{tiling of } \mathcal{A}_k)$ *for $k \geq 2$.*

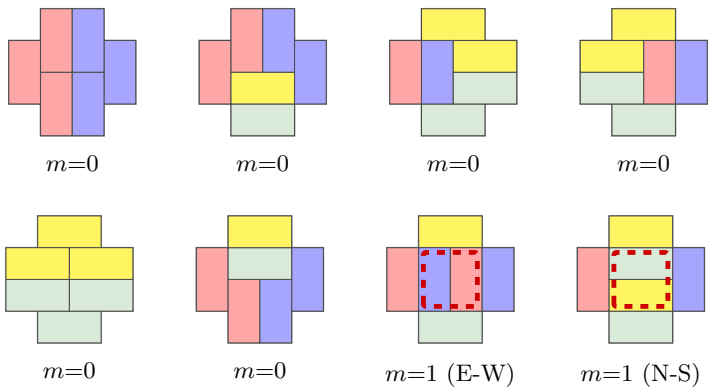
The mechanism is simple: the RSK step always uses exactly $k+1$ random bits, while the shuffling step uses $k+1+m$ bits, where m is the number of bad blocks in the $\mathcal{A}_k$ tiling. When $m > 0$, the shuffling has strictly more randomness in the conditional step.

3 Example: $\mathcal{A}_2 \to \mathcal{A}_3$

The Aztec diamond $\mathcal{A}_2$ has $2^{2 \cdot 3/2} = 8$ tilings. Of these, 6 have no bad blocks ($m = 0$) and 2 have exactly one bad block ($m = 1$).

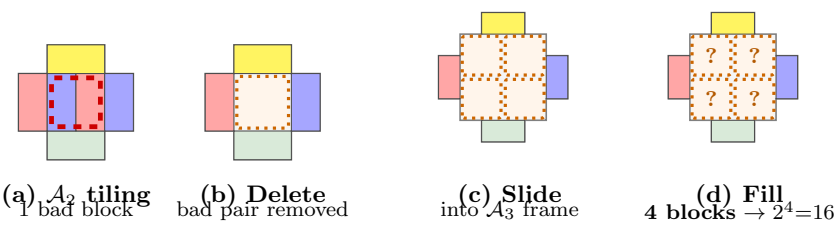
3.1 The eight tilings of $\mathcal{A}_2$

We display all eight tilings below. Colors encode domino types: N (north), S (south), E (east), W (west). Bad blocks are marked with .



3.2 The shuffling step for a bad-block tiling

Consider Tiling 7 (E-W bad block). The shuffling step $\mathcal{A}_2 \to \mathcal{A}_3$ proceeds as follows:



Comparison. For a tiling with $m = 0$ bad blocks, both algorithms use $k+1 = 3$ random bits and produce $2^3 = 8$ equally likely extensions. For a tiling with $m = 1$ bad block:

	EKLP Shuffling	**RSK Growth Diagram**
Random bits per step	$k + 1 + m = 4$	$k + 1 = 3$
Possible extensions	$2^4 = 16$	$2^3 = 8$
Probability per ext.	$\frac{1}{16}$ each	$\frac{1}{8}$ each

In the shuffling, deleting a bad block frees a 2×2 region that later gets re-filled with an independent coin flip, introducing one extra random bit per bad block. Since the RSK boundary layer has no analog of deletion, its conditional distribution uses strictly fewer random choices.

Remark 2. *Despite the different conditional step distributions, both algorithms produce the same* marginal *distribution at every level: uniform over all $2^{n(n+1)/2}$ tilings of $\mathcal{A}_n$. The difference lies entirely in the coupling between consecutive levels.*

4 Empirical verification

We ran 5×10^5 samples per algorithm for the step $\mathcal{A}_2 \to \mathcal{A}_3$. The partition-sequence encoding was used to identify tilings (with a conjugation correction to match conventions between the two algorithms).

Marginal distributions. Both algorithms produced all 8 tilings of $\mathcal{A}_2$ and all 64 tilings of $\mathcal{A}_3$ with the correct uniform frequencies. The total variation distance between the marginal distributions was < 0.002 for $\mathcal{A}_2$ and < 0.007 for $\mathcal{A}_3$, consistent with sampling noise. The chi-squared p-values were 0.57 and 0.44, respectively.

Conditional distributions. For the 6 states with $m = 0$: conditional TV distances were all below 0.01 with chi-squared p-values > 0.06, consistent with identical distributions.

For the 2 states with $m = 1$: conditional TV distances were approximately $\mathbf{0.50}$, with chi-squared p-values effectively zero ($\chi^2 > 40{,}000$). The shuffling produced 16 distinct $\mathcal{A}_3$ extensions from each of these states (each with probability $\approx 1/16$), while RSK produced only 8 (each with probability $\approx 1/8$).

Joint distribution. The joint distribution on $(\mathcal{A}_2, \mathcal{A}_3)$ pairs showed a TV distance of 0.13, with the shuffling producing 80 distinct pairs versus 64 for RSK. The extra 16 pairs correspond to the 2 bad-block states $\times$ 8 extensions that only the shuffling can reach.

5 Discussion

The EKLP shuffling and RSK growth diagram define two different *couplings* between $\mathcal{A}_k$ and $\mathcal{A}_{k+1}$ tilings that share the same marginals but differ in their conditional structure. The shuffling coupling is "more random" due to the delete-and-refill mechanism: each bad block contributes one extra fair coin flip to the transition.

For $k = 1$, neither algorithm produces bad blocks (the unique pair in $\mathcal{A}_1$ is never a colliding pair), so the conditional distributions coincide. Starting from $k = 2$, some $\mathcal{A}_k$ tilings contain bad blocks, and the conditional distributions diverge.

The number of bad blocks m depends on the specific $\mathcal{A}_k$ tiling. For $\mathcal{A}_2$, the maximum is $m = 1$ (attained by 2 of the 8 tilings). In general, m can grow with k.